

## The implication of input data aggregation on up-scaling soil organic carbon changes



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### ABSTRACT

In up-scaling studies, model input data aggregation is a common method to cope with deficient data availability and limit the computational effort. We analyzed model errors due to soil data aggregation for modeled SOC trends. For a region in North West Germany, gridded soil data of spatial resolutions between 1 km and 100 km has been derived by majority selection. This data was used to simulate changes in SOC for a period of 30 years by 7 biogeochemical models. Soil data aggregation strongly affected modeled SOC trends. Prediction errors of simulated SOC changes decreased with increasing spatial resolution of model output. Output data aggregation only marginally reduced differences of model outputs between models indicating that errors caused by deficient model structure are likely to persist even if requirements on the spatial resolution of model outputs are low.

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## 1. Software availability

Gnuplot 5.0; Copyright (C) 1986–1993, 1998, 2004, 2007–2015 Thomas Williams, Colin Kelley and many others; <http://gnuplot.sourceforge.net/>; Hardware: any PC; Software requirement: Gnuplot is a portable command-line driven graphing utility for Linux, OS/2, MS Windows, OSX, VMS, and many other platforms; Free software license. Language: C; Installed size on Linux: 43 kB.

R 3.3.1: A language and environment for statistical computing; Copyright © 1990 W. N. Venables Copyright © 1992 W. N. Venables & D. M. Smith Copyright © 1997 R. Gentleman & R. Ihaka Copyright © 1997, 1998 M. Maechler Copyright © 1999–2016 R Core Team; Copyright (C) 2016 The R Foundation for Statistical Computing; Hardware: any PC; Software requirement: It compiles and runs on a wide variety of UNIX platforms and similar systems (including FreeBSD and Linux), Windows and MacOS. Free Software under the terms of the Free Software Foundation's; Language: C, C++ and FORTRAN; Size: installation package for Windows: 70.4 MB.

CENTURY 4.6; William J. Parton, Dennis Ojima, Rebecca McKeown, and William Pulliam; <http://www.nrel.colostate.edu/projects/century/>; e-mail: [century@colostate.edu](mailto:century@colostate.edu); Hardware: Any PC; Software requirement: Windows or Linux, on Linux system C and FORTRAN compiler; Cost: free Language: C and FORTRAN; Size: 1022 KB.

DailyDayCent; Del Grosso et al., 2001; Parton et al., 1998; Parton et al., 2001; <http://www.nrel.colostate.edu/projects/daycent/>; E-mail: [century@colostate.edu](mailto:century@colostate.edu); Hardware: Any PC; Software requirement: Windows or Linux, on Linux system C and FORTRAN compiler; Cost: free Language: C and FORTRAN; Size: 710 KB.

CoupModel; Per-Erik Jansson et al. & Henrik Eckersten; <http://www.coupmodel.com/>; Contact informations on the website Hardware: Any PC; Software requirement: Windows; Cost: free; Size: 78 MB.

MONIKA 1.2; Claas Nendel & Xenia Specka; <http://monica.agrosystem-models.com/en/download/category/6-installer> Contact informations on the website Hardware: Any PC; Software requirement: Windows; Cost: free; Size: 2.5 MB.

Hermes; K. C. Kersebaum, 1989; [http://www.zalf.de/de/forschung\\_lehre/software\\_downloads/Seiten/default.aspx](http://www.zalf.de/de/forschung_lehre/software_downloads/Seiten/default.aspx) Contact informations on the website; Hardware: Any PC; Software requirement: Windows; Cost: free; Size: 1.1 MB.

STICS; INRA, 1996 <http://www.inra.fr/en/Scientists-Students/Agricultural-systems/All-reports/Modelling-and-agrosystems/STICS-an-agronomy-dynamo>; Contact informations on the website.

APSIM; APSIM Initiative <http://www.apsim.info/>; E-mail: [apsim@daf.qld.gov.au](mailto:apsim@daf.qld.gov.au); Hardware: minimum: Minimum: 1 GHz CPU 512 MB RAM; Recommended hardware: Quad Core 3 GHz + CPU 4 GB RAM; Software requirement: Windows XP/Mono 2.10 for Linux The Microsoft.NET 4 Framework The Microsoft C++ Runtime for Visual Studio 2010; Cost: free but registration required; Language: .NET and C++.

## 2. Data availability

All described crop input data is freely available under the CREATIVE COMMONS ATTRIBUTION 4.0 INTERNATIONAL LICENSE (Hoffmann et al., 2016b).

## 3. Introduction

An important component of soil plant atmosphere systems is the decomposition and sequestration of soil organic matter. SOC content affects almost every process in agro ecosystems. Examples range from soil water budgets (Rawls et al., 2003), the (im-)mobilization of crop nutrients to the exchange of greenhouse gases

between soil and atmosphere (Li et al., 2005). Depending on the respective model structure, model input uncertainties affecting the quantified decomposition and sequestration of SOC will also have implications for modeled crop growth, water balance, nutrient leaching and gaseous losses. Up-scaling studies addressing soil organic carbon (SOC) changes on different scales (Smith et al., 1997, 2005; Lugato et al., 2014; Butterbach-Bahl et al., 2004; Todd-Brown et al., 2013) have been performed to simulate recent and future trends of GHG budgets. Within the scope of national GHG inventories biogeochemical models have been used to improve the estimates of GHG emissions from agro-ecosystems (Del Grosso et al., 2006; EPA, 2016). Considering spatially variable climate, crop management and soil properties as well as their interactions is an important advancement of process-based models as compared to emission factor approaches (Leip et al., 2011; Milne et al., 2007; Smith et al., 2012).

Up-scaling studies often use spatial input data in coarse resolution for two reasons. Either input data on fine resolution is not available or the computational capacity is too low to cope with the large data amounts caused by fine resolution data. The application of low resolution input data introduces uncertainty that interacts with model uncertainty, where the model uncertainty combines parameter uncertainty and model structural uncertainty caused by deficient model sensitivities for the chosen domain and uncertainty caused by unknown or neglected processes (Draper, 1995). Spatial input data aggregation is increasingly used for agro-environmental assessments (Folberth et al., 2016). The quantification of uncertainty caused by input data aggregation in relation to other sources of up-scaling uncertainty is essential to design adequate aggregation schemes and produce reliable up-scaling results. Methods need to be developed that translate observed aggregation effects into ex ante uncertainty estimation methods. Numerous studies analyzed prediction errors caused by spatially aggregated input data (listed in Hoffmann et al., 2016a). It was found that prediction errors are influenced by the type of addressed response variable (Mo et al., 2009; Zhao et al., 2015b), the type of low resolution input data (Kersebaum and Wenkel, 1998; Hoffmann et al., 2016a), the spatial resolution of aggregated data (Hoffmann et al., 2015, 2016a; Zhao et al., 2015b) and the sensitivity of the model (Mearns et al., 1999; Zhao et al., 2015b).

In regional soil carbon modeling studies, the uncertainty of soil properties is an important source of prediction uncertainty. Available maps of soil properties are products of regionalization methods based on soil surveys (Martin et al., 2011; Wiesmeier et al., 2013) or synthesis of soil property databases and classification schemes (Trangmar et al., 1986; Wösten et al., 1999). Often soil mapping units are clustered with one cluster containing the information about the share of main soil types (Jones et al., 2005), but not their spatial distribution within a mapping unit. Thus, up scaling studies addressing SOC changes are prone to prediction uncertainties caused by deficient soil property information. Effects of deficient soil data on modeled crop yields (Easterling et al., 1998; Olesen et al., 2000; Folberth et al., 2012; Angulo et al., 2014; Hoffmann et al., 2016a), net primary productivity (Pierce and Running, 1995) nitrogen leaching (Kersebaum and Wenkel, 1998) or greenhouse gas emissions (Yu et al., 2011) have been examined using spatially aggregated soil data for variable resolutions. Effects of soil data resolution on modeled SOC changes were described by Zhang et al. (2014a,b, 2015a,b) for paddy rice soils of the Tai-Lake region in China using the DNDC model. Zhang et al. (2014a,b) showed that errors of the initial SOC map had the biggest effect on modeling accuracy comparing model results for vectorised soil data in a resolution of 1:50.000 and 1: 14.000.000. However, studies describing the relationship between aggregated soil data variables and aggregation errors are missing.

In up-scaling studies, the required spatial resolution of the model output is a key variable controlling the magnitude of prediction errors arising from input data resolution (Heuvelink, 2002). Therefore, model input data aggregation schemes within up-scaling studies have to be designed with respect to the addressed scale of interest. (Heuvelink and Pebesma, 1999; Heuvelink, 2002). In terms of SOC modeling, SOC stock changes at regional (Ogle et al., 2010), national (EPA, 2016), continental (Smith et al., 2005; Lugato et al., 2014) and global scale (Todd-Brown et al., 2013) have been examined. Also national inventories in the context of international commitments (UNFCCC, Kyoto) report rather spatially aggregated data on SOC stock changes within the sector LULUCF (land use, land use change and forestry).

The spatial aggregation of model results can reduce errors from deficient soil input data and model structure (including parameter uncertainty) because local over- or underestimations caused by deficient input data or not correctly defined model sensitivities might level out by spatial aggregation (Heuvelink, 2002; Leip et al., 2011).

Ogle et al. (2010) predicted SOC trends of US croplands with CENTURY and quantified uncertainties caused by model structure and uncertain input data. Shifting the scope of interest from point scale to regional and country scale, clearly reduced median uncertainties (half of the 95% confidence interval divided by the mean for SOC changes) from 739% to 674% to 114–118% for the regional scale and to 16–22% for the country scale, respectively. Model structural uncertainty contributed with around 97% to overall uncertainty at the national scale. This shows that the relative contribution of uncertainty from soil data and model structure varies depending on the spatial resolution of analyzed up-scaling results. Unfortunately, Ogle et al. (2010) did not exclusively address soil aggregation effects in their uncertainty analyses.

Soil data aggregation affects modeled SOC changes by several mechanisms. First the local initial SOC is compromised. Foreid et al. (2012) reported strong effects of modified initial SOC stocks on modeled SOC changes for England and Wales quantified with DayCent compared to effects of different climate scenarios. Second, crop growth and therewith the amount of crop residues entering the soil is affected by soil data aggregation. Olesen et al. (2000) and Hoffmann et al. (2016a) found significant effects of soil data aggregation on crop yields but did not examine carbon input. Moreover the dynamics of decomposition are affected by soil properties either in a direct way or indirectly via soil temperature, water content or nutrient availability.

The goal of this study is to analyze the effect of soil input data aggregation on modeled SOC stocks and changes in dependence on spatial resolution of soil data, the used model and the spatial resolution of output data and time. We assume that aggregation effects will be largely controlled by the aggregation error of initial SOC maps (Zhang et al., 2014a,b). Therefore generic approaches that translate aggregation effects of initial SOC maps into aggregation effects of modeled SOC stocks are developed and tested. According to findings of Ogle et al. (2010) we hypothesize that errors of coarse resolution model results are dominated by model structural errors. We analyze the variability between models in dependence on the model output resolution to compare upscaling effects caused by data aggregation and model algorithms.

## 4. Material and method

### 4.1. Study region

The study region was North Rhine-Westphalia (NRW, 6 East–9.5 East, 50 North–52.5 North) in the West of Germany with an area of 34098 km<sup>2</sup>. The highest point of the area is 843 m above sea level.

Approximately the 60% of the area is used as agricultural land and the dominant crops based on the report of the Federal Statistical Office (2013) are silage maize and winter wheat. The whole area of North Rhine-Westphalia was treated as cropland (Zhao et al., 2015a).

### 4.2. Model input data

#### 4.2.1. Weather data

A fine resolution gridded data set generated by combining gridded data in daily to monthly resolution and daily station measurements of 280 weather stations from the German Meteorological Service (DWD, 2012) was used within this modeling exercise. Meteorological data sets have a resolution of 1 km and comprise daily air temperature, precipitation, global radiation, wind speed and relative humidity within the time period 1981–2011. A detailed description on data processing can be found in Zhao et al. (2015a) while Hoffmann et al. (2015) provide statistical analyses of this data set.

#### 4.2.2. Soil data

Gridded reference data on soil properties had the same extent and resolution like the weather data and is based on the soil map of the Geological Service North Rhine-Westphalia (1:50,000) (Geological Service NRW, 2004; Angulo et al., 2014; AG-Boden, 2005). In this map, each soil type is characterized by a unique set of parameters defining texture, soil hydraulic and chemical properties. Missing information on SOC content and C:N ratios was estimated by pedotransfer functions (Angulo et al., 2014; AG Boden, 1994). SOC content and pH came from the FIS StoBo (LANUV, 2014). Gravel content and the bulk density corrected for gravel content were estimated by pedotransfer functions (Poesen and Lavee, 1994).

#### 4.2.3. Input data aggregation

Soil data was aggregated to raster maps of  $1 \times 1$ ,  $10 \times 10$ ,  $25 \times 25$ ,  $50 \times 50$  and  $100 \times 100$  km x km grid cell size. A unique soil data set representing the region of North Rhine-Westphalia was generated. The dominant soil type within an aggregation unit was chosen as aggregation criterion. Climate data was kept at 1 km resolution.

#### 4.2.4. Management practice

Model runs for the period 1981 to 2011 were performed for maize and winter wheat monocultures. Cropping periods were from October 1 to August 1 for winter wheat and from April 20 to September 20 for maize. Annually, winter wheat received mineral fertilizer of 208 kg N ha<sup>-1</sup> in three rates from March to June, while maize was fertilized by 238 kg N ha<sup>-1</sup> in April (30 kg N ha<sup>-1</sup>) and June (208 kg N ha<sup>-1</sup>). At harvest, above ground crop residues were removed for both cropping systems.

### 4.3. Participating models and model scenarios

SOC sequestration was modeled by 7 models, namely Daily-DayCent, MONICA, HERMES, COUP, STICS, APSIM and CENTURY (supplementary material A). Beside the modeled annual SOC of the top soil (30 cm) at seeding time, annual yields, net primary production (NPP), net ecosystem exchange (NEE) and aboveground biomass at harvest (AGB) were recorded per model run. Models were trained to reproduce mean yields of winter wheat and silage maize for NRW. No further calibration procedures were conducted. All models had been evaluated on long term data of SOC stocks, organic nitrogen or CO<sub>2</sub> emissions in temperate climate (Model description in Supplementary material A).

For all participating models the description of SOC cycling is based on pool concepts. Pools characterized by specific decomposition rates ( $k$ ) are connected by fluxes of decomposition products. SOC decomposition is accompanied by  $\text{CO}_2$  evolution. The fraction of  $\text{CO}_2$  losses from decomposition products are functions of the carbon use efficiency. The pool specific decomposition of SOC is described by first order kinetics (equation (1)).

$$\frac{d\text{SOC}}{dt} = K(T, \theta, \dots) \cdot \text{SOC} \quad (1)$$

Pool specific decomposition rates are modified by soil moisture ( $\theta$ ), temperature ( $T$ ), texture and for some models by nitrogen availability (DailyDayCent, CENTURY, STICS). Six models implement an inert (APSIM, STICS, HERMES, MONICA) or quasi inert pool (passive pool in DailyDayCent and Century) while the COUP does not. Carbon use efficiencies are fixed or determined by soil texture (DailyDayCent, Century). Although also based on pool concepts, the models HERMES and STICS describe soil organic matter decomposition based on nitrogen and use flexible or fixed C:N ratios to determine SOC fluxes. For the majority of models, the initial distribution of soil organic matter (SOM) between pools was based on fixed fractions a priori defined in relation to texture, land use or initial SOC. DailyDayCent scenario runs followed on a priori spin-up run driven by the local climate conditions and the assumed management (maize and winter wheat cropping).

#### 4.3.1. Description of the aggregation effect

The effect of spatial soil data aggregation depends on the scale of soil input aggregation and on the addressed output aggregation scale. To analyze the relationships between soil data aggregation and required output resolution three types of scales were defined:

1. The **reference scale** (abbreviation: ref): which represents the 1 km high resolution. This is also the resolution at which simulations were performed (computational scale).
2. The **aggregation scale** (abbreviation: aggr): which stands for the aggregation level of soil data aggregation (10, 25, 50, 100 km and the whole region of North Rhine-Westphalia [NRW])
3. The **scale of interest** (or model output aggregation scale) which is the aggregation level where information on modeled SOC stocks or changes is required (1, 10, 25, 50, 100 km and NRW the region of North Rhine-Westphalia)

The variables addressed in this analysis were the top soil (30 cm) SOC at time  $t$  ( $\text{SOC}_t$ ) and the SOC change ( $\Delta\text{SOC}_t$ ) between  $\text{SOC}_0$  (SOC at time 0) and  $\text{SOC}_t$ :

$$\Delta\text{SOC}_t = \text{SOC}_0 - \text{SOC}_t \quad (2)$$

The model error of a  $1 \times 1$  km x km grid cell ( $\Delta E_{\text{SOC}_t}$  and  $\Delta E_{\Delta\text{SOC}_t}$ ) is then calculated by the difference between the target variable of the reference scale (ref) and the target variable of the aggregation scale (aggr) (see Fig. 1a):

$$\Delta E_{\text{SOC}_t} = \text{SOC}_{t \text{ ref}} - \text{SOC}_{t \text{ aggr}} \quad (3)$$

$$\Delta E_{\Delta\text{SOC}_t} = \Delta\text{SOC}_{t \text{ ref}} - \Delta\text{SOC}_{t \text{ aggr}} \quad (4)$$

The root mean squared error was used to assess the model error from soil input data aggregation:

$$\text{RMSE}(\text{SOC}_t) = \sqrt{\frac{\sum_n (\Delta E_{\text{SOC}_t})^2}{n}} \quad (5)$$

$$\text{RMSE}(\Delta\text{SOC}_t) = \sqrt{\frac{\sum_n (\Delta E_{\Delta\text{SOC}_t})^2}{n}} \quad (6)$$

here  $n$  is the number of  $1 \text{ km} \times 1 \text{ km}$  grid cells (computational scale). To relate aggregation effects to modeled SOC changes the RMSE was divided by the mean absolute SOC change of the models at reference scale

$$r\text{RMSE}(\Delta\text{SOC}_t) = \frac{\text{RMSE}(\Delta\text{SOC}_t)}{E|\Delta\text{SOC}_t|} \quad (7)$$

where  $E()$  is the expectation. To evaluate aggregated model output at coarse resolution (scale of interest: 10 km, 25 km, 50 km, 100 km, NRW) model results of the input aggregation scale and the reference scale were aggregated by averaging. These averaged values were used to calculate the RMSE (Fig. 1b). In this case,  $\Delta E_{\Delta\text{SOC}_t}$  for a grid cell corresponding to the scale of interest results from:

$$\overline{\Delta E_{\Delta\text{SOC}_t}} = \left( \frac{\sum_{i=0}^m \Delta\text{SOC}_{i \text{ ref}}}{m} \right) - \left( \frac{\sum_{i=0}^m \Delta\text{SOC}_{i \text{ aggr}}}{m} \right) \quad (8)$$

here  $m$  is the number of  $1 \text{ km} \times 1 \text{ km}$  grid cells within an element corresponding to the scale of interest ( $\overline{\Delta E_{\Delta\text{SOC}_t}}$  is quantified accordingly).

#### 4.3.2. Description of simulation error caused by model variability

The effect of the model error on modeled SOC development cannot be quantified because the real temporal response of SOC on assumed management and climate are unknown. However, the variability between different models is one component of model ensemble error. Maiorano et al. (2016) used the mean squared error (MSE) to describe model ensemble error. The mean squared error results from the variance (var) and bias of the model ensemble, respectively:

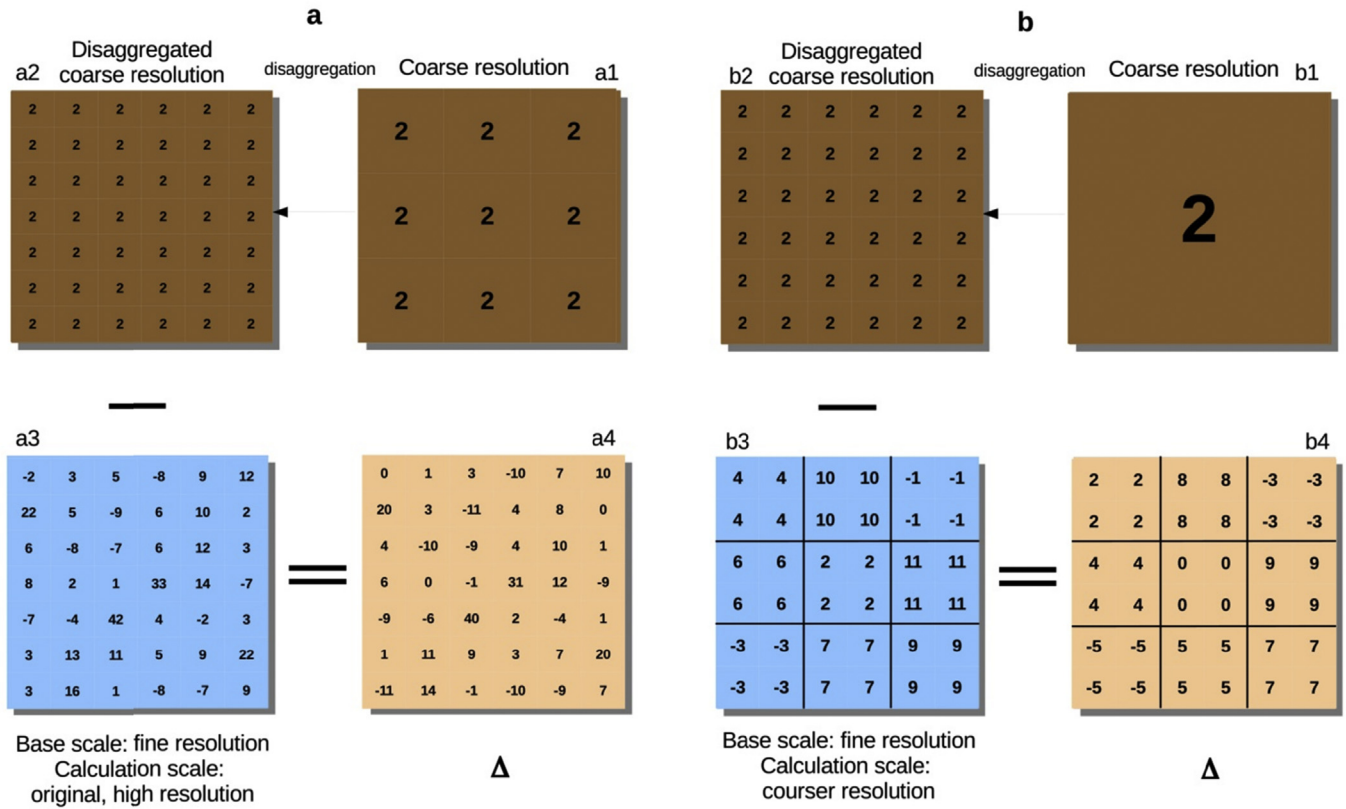
$$\begin{aligned} \text{MSE}_{\text{MME}} &= \frac{1}{n} \frac{1}{g} \sum_{n_i=1}^n \sum_{g_i=1}^g (y - \hat{y}_{n_i, g_i})^2 \\ &= \frac{1}{n} \sum_{n_i=1}^n \text{var}_g(y_g) + \frac{1}{n} \sum_{n_i=1}^n (\text{bias}(y_{n, g}))^2 \end{aligned} \quad (9)$$

here  $n$  is the number of grid cells of the reference grid,  $g$  is the number of participating models,  $y$  represents observations and  $\hat{y}$  the simulated values, respectively. While the bias depends on observations and simulations the variance only takes into account simulated values (Maiorano et al., 2016). Due to the lack of observations we restrict the description of effects caused by model structural errors to the variance of the model ensemble, knowing well that this formulation represents the lower limit of the model related error (bias and var related summands of equation (9) are always positive). When an unbiased model ensemble is assumed the corresponding RMSE of the model ensemble for different levels of output data aggregation is:

$$\text{RMSE}_{\text{ensemble}} = \sqrt{\frac{1}{n/m} \sum_{nm_i=1}^{n/m} \text{var}(y_{g, nm_i})} \quad (10)$$

here  $n$  is the number of grid cells of the reference grid and  $m$  is the number of reference grid cells within a grid cell of the output aggregation grid (scale of interest). Effects of model variability were quantified only for the reference scale and then aggregated according to scales of interest (1 km, 10 km, 25 km, 50 km, 100 km and





**Fig. 1.** (a) Scheme of the RMSE calculation (scale of interest: fine resolution) with the model output for coarse resolution input data (a1,b1), the spatial disaggregated model output (a2,b2), the model results for fine resolution input data (a3, spatially aggregated for b3) and the difference of model results from coarse and fine resolution input data used for the RMSE quantification and (b) the scheme of the RMSE calculation (scale of interest: coarser resolution).

NRW). The model results of DailyDayCent were not used for the computation of model ensemble means because DailyDayCent runs are based on spin-up runs and are therefore, not comparable to the remaining models.

#### 4.4. Aggregation effects based on a simple soil carbon model

We adopted the concept of first order kinetics shared by the majority of participating models to formulate a Simple soil CARbon model (SICA). SICA was used to examine the effects of model driving data (SOC initialization, carbon input from crop residues), model parameters and their aggregation errors on the temporal dynamics of aggregation effects. SICA was enabled to represent decomposition dynamics and sensitivities on driving factors (initial SOC, carbon input) of participating biogeochemical models by calibration on respective model output data. The interaction of model driving data, their errors and computed SOC stocks and changes were analyzed first at the smallest computational unit and then scaled up to the study region.

##### 4.4.1. The simple soil carbon model SICA

Compared to dynamics of other biogeochemical processes like crop growth or nutrient leaching the sequestration and decomposition of SOC are slow processes. Often it takes decades to observe significant changes of SOC stocks (Smith, 2004) and up-scaling studies rather address trends in SOC development than annual fluctuations (Smith et al., 2005; Foreid et al., 2012; Lugato et al., 2014). At intermediate time scales the effect of inter annual variability of model driving data on SOC stocks diminishes and the model error caused by temporal aggregated driving data is negligible. Therefore, we applied the analytical solution of SICA driven

by temporally aggregated carbon input from crop residues (crop biomass entering the soil from crop residues and root exudation).

SICA contains two C-pools, one representing the decomposable SOC and the other representing the inert pool (Fig. 2).

The related differential equation for the model depicted in Fig. 2 is:

$$\frac{dSOC}{dt} = (k(t) \times (SOC(t) - IOM) + R(t)) \quad (11)$$

with  $R$  [t C ha<sup>-1</sup> a<sup>-1</sup>] the C input rate,  $SOC$  [t C ha<sup>-1</sup>] the SOC concentration,  $IOM$  [t C ha<sup>-1</sup>] the inert carbon pool, and  $k$  the decomposition rate ( $K$ ) multiplied by  $-1$  ( $k = -K$ ) [a<sup>-1</sup>] and  $t$  [a] as time. Assuming that the modeling period is large enough  $R$  and  $k$  could be represented by some averaged (time-invariant) carbon input and decomposition rate. Further, it is assumed that for model initialization the IOM is a constant fraction of the SOC.

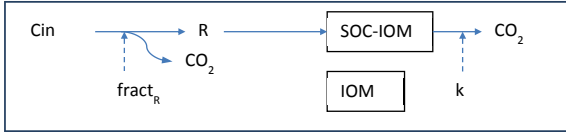
$$IOM = fract_{IOM} SOC_0 \quad (12)$$

Solving equation (11) (see supplementary material B) results in:

$$SOC_t = \left( SOC_0 (1 - fract_{IOM}) + \frac{R}{k} \right) \exp(kt) - \frac{R}{k} + SOC_0 fract_{IOM} \quad (13)$$

##### 4.4.2. Aggregation effects for a single grid cell

For the majority of applied models, the dynamics of SOC decay is determined by the interaction of decomposition rates, carbon use efficiencies and the distribution of SOC between C-pools. In the generic approach, the decomposition rate  $k$  implicitly accounts for



**Fig. 2.** Generic SOC model; SOC = soil organic carbon stock; CO<sub>2</sub> = Emitted CO<sub>2</sub>; IOM = inert SOC pool; R = input rate into the SOC pool; Cin = carbon input of crop residues and organic amendments;  $fract_R$  = IOM/SOC<sub>0</sub>; K = decomposition rate.

these interactions. For the majority of applied models soil properties affect carbon use efficiencies and/or decomposition rates. For instance, in CENTURY the partition of decomposed SOC from the active pool into CO<sub>2</sub> loss and sequestered SOC is driven by the sand content (Parton et al., 1992). Soil properties (e.g. texture, bulk density, available water content, etc.) also control the soil temperature and the soil moisture both known to affect SOC decomposition rates. These interactions are considered in applied models but implemented in different ways. The modification of soil properties of a single computational grid cell by soil data aggregation will affect decomposition rates and carbon use efficiencies of participating models and will therefore influence the decomposition dynamics. In SICA these are represented by the decomposition rate  $k$ .

Another variable influenced by soil data aggregation is the carbon input from crop residues. Soil properties drive soil water dynamics and plant available nutrients in participating models and thus the spatial and temporal pattern of modeled crop development is affected by soil data aggregation as shown in Hoffmann et al. (2016a) for crop yields.

Following these considerations error terms were defined for the initial SOC (SOC at time  $t = 0$ ), the temporal averaged decomposition rate  $K$  and carbon input  $R$ :

$$k_{ref} = k_{agg} + \epsilon_k \quad (14)$$

$$SOC_{ref} = SOC_{agg} + \epsilon_{SOC} \quad (15)$$

$$R_{ref} = R_{agg} + \epsilon_R \quad (16)$$

here the subscripts *ref* and *agg* refers to the reference scale and the aggregation scale, respectively. The combination of equations (3) and (4) and (13)–(16) results in equations that describe the error of SOC<sub>t</sub> and ΔSOC<sub>t</sub> by data aggregation per grid cell (see supplementary material D and E for the derivation):

$$\begin{aligned} \Delta E_{SOC_t} = & SOC_{agg}(1 - fract_{IOM})\exp(kt)(\exp(\epsilon_k t) - 1) \\ & + \epsilon_{SOC_0}((1 - fract_{IOM})\exp(kt)\exp(\epsilon_k t) + fract_{IOM}) \\ & + R\left(\frac{\exp((k + \epsilon_k)t) - 1}{k + \epsilon_k} - \frac{(\exp(kt) - 1)}{k}\right) \\ & + \epsilon_R\frac{(\exp((k + \epsilon_k)t) - 1)}{k + \epsilon_k} \end{aligned} \quad (17)$$

$$\begin{aligned} \Delta E_{\Delta SOC_t} = & SOC_{agg}(1 - fract_{IOM})\exp(kt)(\exp(\epsilon_k t) - 1) \\ & + \epsilon_{SOC_0}(1 - fract_{IOM})(\exp(kt)\exp(\epsilon_k t) - 1) \\ & + R\left(\frac{\exp((k + \epsilon_k)t) - 1}{k + \epsilon_k} - \frac{(\exp(kt) - 1)}{k}\right) \\ & + \epsilon_R\frac{(\exp((k + \epsilon_k)t) - 1)}{k + \epsilon_k} \end{aligned} \quad (18)$$

From equations (17) and (18) it follows that the model error per cell caused by spatial aggregation are linearly dependent on SOC<sub>agg</sub>,

ΔSOC<sub>0</sub>,  $R_{agg}$  and  $\Delta R$  but nonlinear on  $k$  and  $\Delta k$ . The aggregation effect per cell is time dependent. The absolute value of SOC will influence the temporal course of the aggregation effect. This is demonstrated by the initial SOC at 60, 230 and 30 t C ha<sup>-1</sup> (see Tab F.1 for parameter settings). The difference between original SOC and aggregated SOC is always 30 t C ha<sup>-1</sup>. The original decomposition rate is set to 0.02 while the decomposition rate influenced by aggregation is set to 0.03 (see Fig. F1 for modeled SOC time series driven by original and aggregated input data). Before aggregation effects and normalized aggregation effects (divided by initial ΔSOC) for the three cases become similar near to equilibrium they show a rather high variability (Fig. 3). This demonstrates that if decomposition rates are affected, the aggregation error depends not only on differences between original and aggregated data but also on absolute amounts of SOC and  $R$ .

Equations (17) and (18) can be simplified for special cases:

#### Case 1. Equilibrium.

If the length of the simulation period is adequately long and the model turns to be in equilibrium, the terms  $\exp(kt)$  and  $\exp(\epsilon_k t)$  will become zero resulting in simplifications of equations (17) and (18).

$$\Delta E_{SOC_t} = \epsilon_{SOC_0} fract_{IOM} + R\left(\frac{-1}{k + \epsilon_k} - \frac{-1}{k}\right) + \epsilon_R \frac{-1}{k + \epsilon_k} \quad (19)$$

$$\Delta E_{\Delta SOC_t} = -\epsilon_{SOC_0}(1 - fract_{IOM}) + R\left(\frac{-1}{k + \epsilon_k} - \frac{-1}{k}\right) + \epsilon_R \frac{-1}{k + \epsilon_k} \quad (20)$$

In this case the aggregation effect on SOC and ΔSOC (ΔE<sub>SOC<sub>t</sub></sub>; ΔE<sub>ΔSOC<sub>t</sub></sub>) is just determined by the errors of initial SOC, the inert fraction and errors of assumed decomposition rate and carbon input. The initial SOC is not influencing the aggregation effect. The effect of  $\epsilon_R$  will be stronger the closer  $k$  is approaching zero. Conditions favoring decomposition will provoke a smaller aggregation effect than conditions hampering soil carbon decomposition. Model errors caused by deficient initial SOC values do not interact with deficient decomposition dynamics ( $\epsilon_k$ ) but deficient carbon input rates ( $\epsilon_R$ ) do which is indicated by the last summands of equations (19) and (20).

#### Case 2. Soil data aggregation does not affect the decomposition rate.

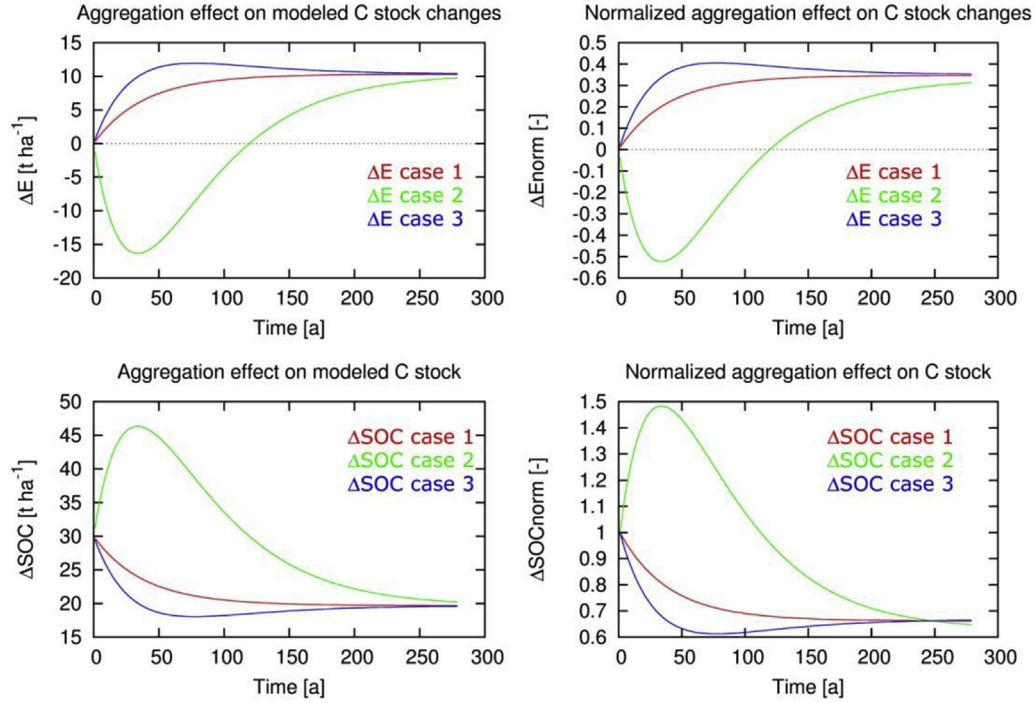
Identical decomposition rates  $K_{ref}$  ( $-k_{ref}$ ) and ( $k_{agg}$ ) will simplify equations (17) and (18) to equations (21) and (22):

$$\begin{aligned} \Delta E_{SOC_t} = & \epsilon_{SOC_0}((1 - fract_{IOM})\exp(kt) + fract_{IOM}) \\ & + \epsilon_R \frac{(\exp(kt) - 1)}{k} \end{aligned} \quad (21)$$

$$\Delta E_{\Delta SOC_t} = \epsilon_{SOC_0}(1 - fract_{IOM})(\exp(kt) - 1) + \epsilon_R \frac{(\exp(kt) - 1)}{k} \quad (22)$$

here  $\epsilon_{SOC_0}$  and  $\epsilon_R$  stand for the differences of SOC and  $R$  between modeling with original input data and aggregated data. These differences depend on the variability of the original data, the chosen scale and method of data aggregation. SOC and  $R$  itself do not determine the aggregation effect. The aggregation effect depends on the simulation period.

#### Case 3. Aggregation does not affect the decomposition rate and the carbon input rate.



**Fig. 3.** Aggregation effects ( $\Delta E$ ,  $\Delta SOC$ ) and normalized aggregation effects ( $\Delta E_{SOC_t}/\epsilon_{SOC_0}$ ,  $\Delta E_{SOC_t}/\epsilon_{SOC_0}$ ) with deficient decomposition kinetics ( $\epsilon_k > 0$ ) and initial SOC ( $\epsilon_{SOC}$  is always  $-30 \text{ t ha}^{-1}$ ). Initial SOC at reference scale ( $SOC_{ref}$ ) is  $60 \text{ t ha}^{-1}$  (case 1),  $250 \text{ t ha}^{-1}$  (case 2) and  $30 \text{ t ha}^{-1}$  (case 3).

For the assumed conditions equations (17) and (18) become:

$$\Delta E_{SOC_t} = \epsilon_{SOC_0}((1 - fract_{IOM}) \exp(kt) + fract_{IOM}) \quad (23)$$

$$\Delta E_{\Delta SOC_t} = \epsilon_{SOC_0}(1 - fract_{IOM})(\exp(kt) - 1) \quad (24)$$

Normalizing equations (23) and (24) by  $\Delta SOC$  results in:

$$\frac{\Delta E_{SOC_t}}{\epsilon_{SOC_0}} = (1 - fract_{IOM}) \exp(kt) + fract_{IOM} \quad (25)$$

$$\frac{\Delta E_{\Delta SOC_t}}{\epsilon_{SOC_0}} = (1 - fract_{IOM})(\exp(kt) - 1) \quad (26)$$

If the influence of data aggregation on  $R$  and  $k$  is small the aggregation effect is determined by  $\epsilon_{SOC_0}$ ,  $k$ ,  $fract_{IOM}$  and  $t$  (equations (23)–(26)). The amount of SOC and  $R$  are not relevant for predicting the aggregation effect. Again, this is demonstrated by 3 examples. Here the first and second example just diverges in the assumed amount of original carbon stocks. Distances between initial C stocks are identical, while in the third example the initial C stock distance differs from example 1 and 2 (see Table G1 for model parameters). According to equations (23) and (24) model errors caused by aggregation are similar for example 1 and 2 (Fig. 4). For all examples,

the rate constants do not differ between the examples and the relative change of the total active SOC stock over time is therefore only a function of time.

#### 4.4.3. Aggregation effects of up-scaling

Equations (17) and (18) quantify aggregation effects for a single grid cell. These equations have to be transformed to approximate large scale aggregation effects. Substitution of factors related to  $SOC_{agg}$ ,  $\epsilon_R$ ,  $R$  and  $\epsilon_{SOC_0}$  with  $a_1$ ,  $a_2$ ,  $a_3$ ,  $a_4$  and the variables  $SOC_{agg}$ ,  $\epsilon_{SOC_0}$ ,  $R$ ,  $\epsilon_R$  by F1, F2, F3, F4 gives:

$$\Delta E = a_1 F_1 + a_2 F_2 + a_3 F_3 + a_4 F_4 \quad (27)$$

where  $\Delta E$  is a substitute for  $\Delta E_{SOC_t}$  or  $\Delta E_{\Delta SOC_t}$ . The RMSE of  $\Delta E$  is derived from:

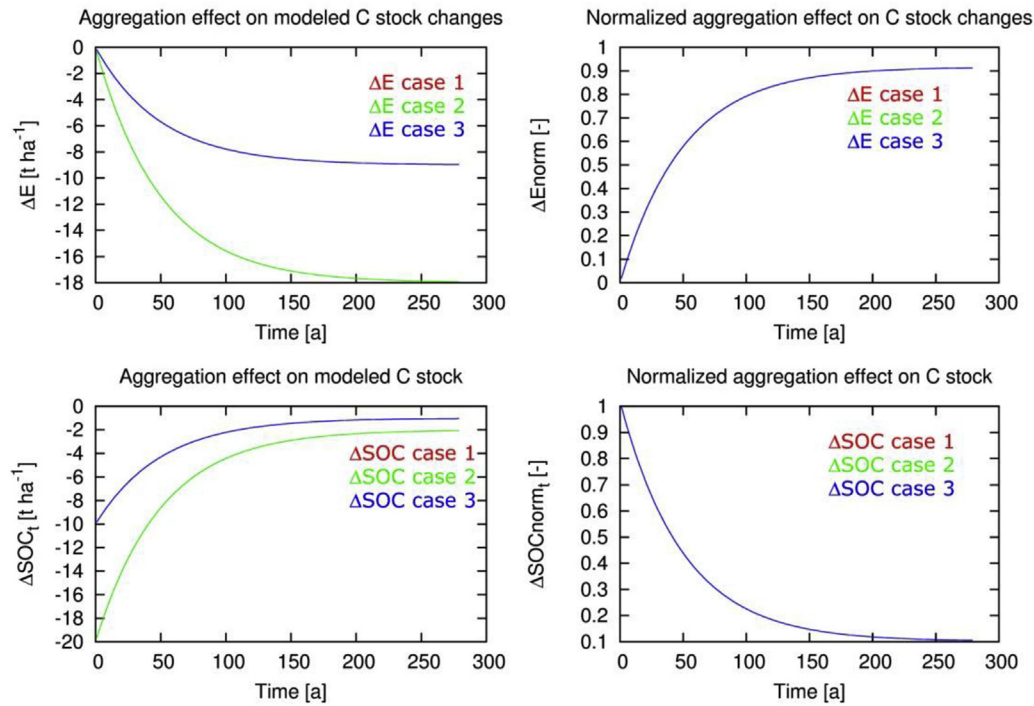
$$RMSE(X) = \sqrt{\frac{\sum_{m=1}^n \sum_{i=1}^4 \sum_{k=1}^4 a_{i-m} F_{i-m} a_{k-m} F_{k-m}}{n}} \quad (28)$$

with  $n$  the number of computational grid cells and  $X$  the modeled variable (either  $SOC_t$  or  $\Delta SOC_t$ ). Assuming that the factors are not independent (correlation exists) the RMSE of  $SOC_t$  and  $\Delta SOC_t$  results from:

$$RMSE(X) = \sqrt{\sum_{i=1}^4 \sum_{k=1}^4 (cov(a_i a_k) + E(a_i)E(a_k))(cov(F_i F_k) + E(F_i)E(F_k)) + cov(a_i a_k, F_i F_k)} \quad (29)$$

normalized model errors are equal under the assumptions of equations (25) and (26) (Fig. 4), which could be expected because

With  $E()$  standing for the expectation and  $cov()$  for the covariance. The occurrence of correlating factors does imply that the



**Fig. 4.** Aggregation effects ( $\Delta E_{\Delta SOC}$ ,  $\Delta E_{SOC}$ ) and normalized aggregation effects ( $\Delta E_{\Delta SOC}/\epsilon_{SOC}$ ,  $\Delta E_{SOC}/\epsilon_{SOC}$ ) assuming  $\epsilon_k = 0$  with initial SOC at reference scale ( $SOC_{ref}$ ) and aggregation effects on initial  $SOC(\epsilon_{SOC})$  of 100 t ha and -20 t ha (case 1), 20 t ha and -20 t ha (case 2), 60 t ha and -10 t ha (case 3).

description of factor uncertainties (a1-a4; F1–F4) alone is not sufficient to estimate aggregation effects and the covariance has to be known for reliable estimates of the RMSE.

For simplified quantification methods (case 3 in section 4.4.2, equations (22) and (23)) the covariance might be negligible because  $\epsilon_{SOC_0}$  is strongly driven by the aggregation itself while  $k$  is independent from soil data aggregation by definition. The fraction of inert  $SOC(fract_{IOM})$  is a model specific constant. The assumptions of case 3 (section 4.4.2) combined with the assumption that  $\Delta SOC$  and the related factor are uncorrelated would result in (derivation in supplementary material H):

$$RMSE(SOC_t) = \sqrt{E(((1 - fract_{IOM}) \exp(kt) + fract_{IOM})^2)} \times RMSE(SOC_0) \quad (30)$$

$$RMSE(\Delta SOC_t) = \sqrt{E(((1 - fract_{IOM})(\exp(kt) - 1))^2)} \times RMSE(SOC_0) \quad (31)$$

where  $E()$  is the expectation.

#### 4.4.4. Calibration of SICA

For each individual model of the model ensemble, we determined the parameters of SICA by calibration on modeled SOC time series. Calibrations were conducted in two steps. First, model parameters  $fract_{IOM}$ ,  $fract_R$  and  $k$  were approximated using the whole data set of SOC time series produced per model. In a second step a single parameter representing the decomposition rate was fine-tuned per grid cell. A quasi Newton method (Byrd et al., 1995) was used for model calibration with the Nash Sutcliffe index (Nash and Sutcliffe, 1970) as optimization criterion. Calibrations of SICA were performed for the aggregation scales and the reference scale, respectively. Factors  $\epsilon_k$  and  $\epsilon_R$  per grid cell were calculated from the

difference of reference scale and aggregation scale.

## 5. Results

### 5.1. Properties of aggregated soil data

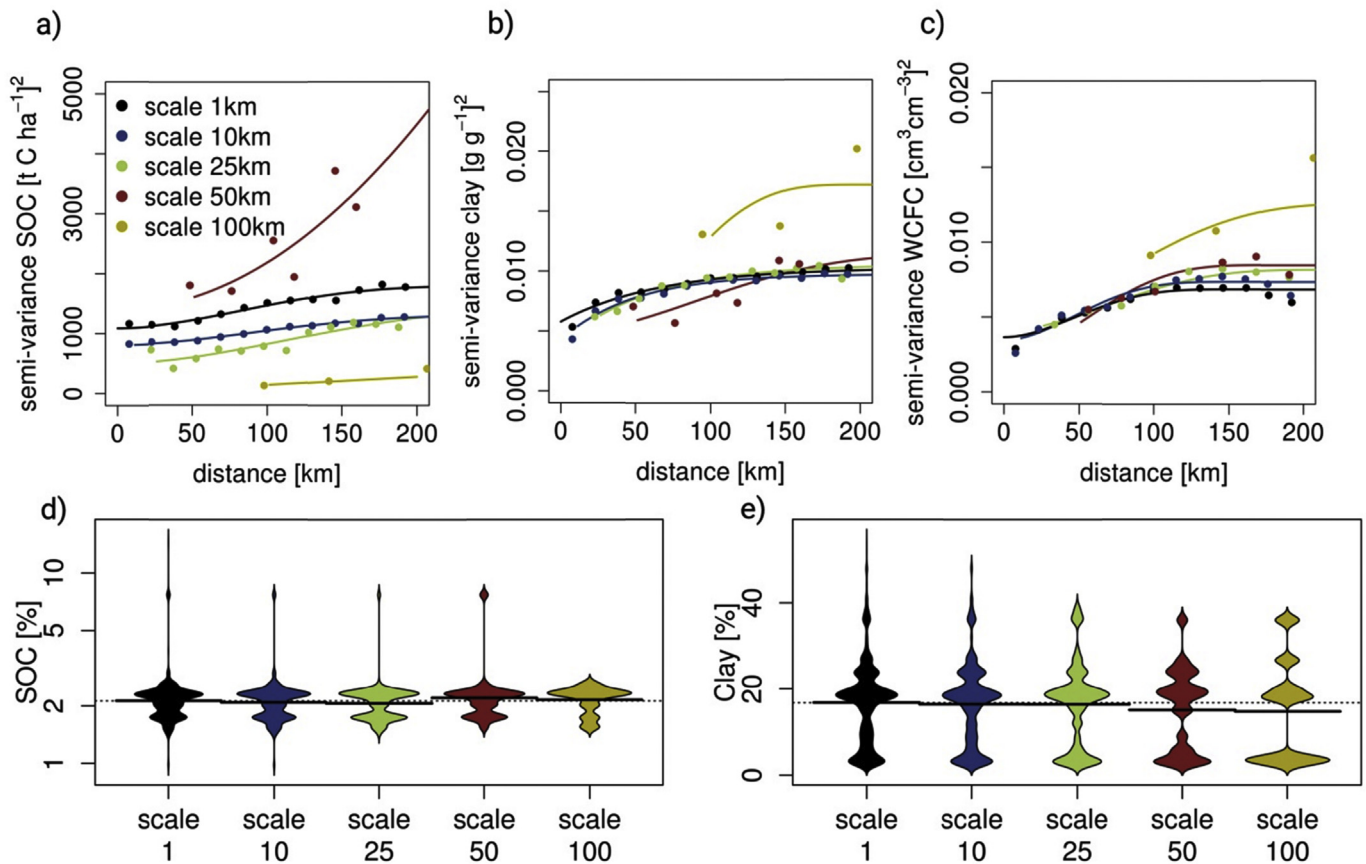
Soil properties at aggregation level and at reference scale equal for the majority of computational grid cells due to the chosen method of aggregation. The share of computational grid cells with similar soil properties between reference and aggregation scale decreases with increasing aggregation level. Soil data aggregation had distinct effects on the spatial variability of soil properties in dependence on the soil property chosen.

Spatial autocorrelation was low for the reference SOC data. Soil data aggregation had only small effects on mean SOC contents and stocks (Fig. 5a). Extreme values of SOC were smoothed by soil data aggregation but aggregation by dominance did not consistently decrease SOC variability (Fig. 5e). While the variability decreased for aggregation levels 10, 25 and 100 km it increased for aggregation level 50 km showing that regional characteristics can be overemphasized by the applied aggregation method. The range of spatial autocorrelation was above the aggregation scale of 100 km. In contrast, soil data aggregation had only marginal effects on the variability of other soil properties like field capacity or clay content for aggregation levels 1 km–50 km. However, spatial variability increased drastically when aggregating to 100 km resolution (Fig. 5 b,c).

### 5.2. Results of model scenarios

With exception of the DailyDayCent model runs, all models predicted SOC losses for continuous winter wheat (Fig. 6) and maize in average indicating that initial soil carbon stocks were far from equilibrium for the management assumed and the assumed spatial initial SOC distribution of the reference scale ( $1 \times 1$  km).





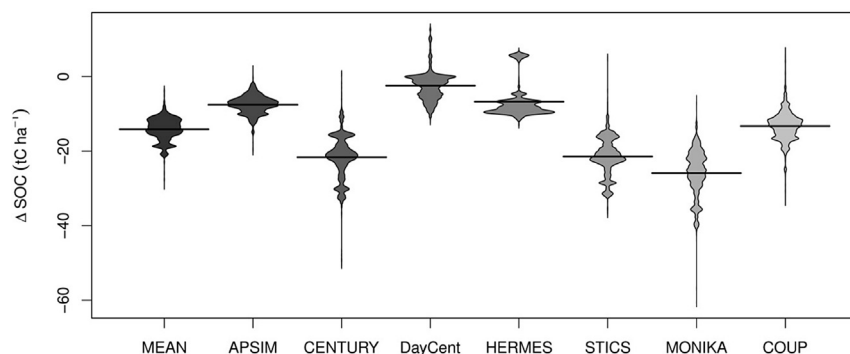
**Fig. 5.** Semi-variograms for aggregated soil properties a) initial SOC, b) clay content, c) field capacity; Violin plots of soil properties (with black lines representing the mean) d) Estimated probability density of SOC [%] and e) Estimated probability density of clay content [%].

Spatial averaged annual SOC losses ranged from 0.1 to 0.75  $\text{t C ha}^{-1}$  with a mean of 0.52  $\text{t C ha}^{-1}$  for wheat and from 0.22 to 0.84  $\text{t C ha}^{-1}$  with a mean of 0.48  $\text{t C ha}^{-1}$  for maize, respectively. Starting from similar values at  $t = 0$ , the variability of spatial averaged SOC stocks between models increased with time.

The spatial pattern of modeled SOC changes was consistent among the models using the given SOC data for model initialization and it was also consistent for the two cropping scenarios. In general, SOC losses were higher in the lowlands of the northwest and smaller in the colder hill ranges in the south and east of North Rhine-Westphalia (Fig. 7). This pattern was consistent with the spatial distribution of mean annual temperature (Zhao et al., 2015b)

indicating that decomposition was delayed in the hilly region with longer periods until equilibrium is reached. In contrast, high decomposition kinetics driven by higher temperature caused fast SOC degradation and an earlier occurrence of equilibrium conditions in the western lowlands.

All models used similar model driving and initialization data the quantified changes in C stocks. However, results varied considerably with regard to averaged SOC changes and spatial variability. In general, high SOC losses were accompanied by a high variability of SOC stock changes and vice versa (Fig. 6). Spatially averaged  $\Delta\text{SOC}_{29}$  (time = 29 a) and  $\text{SOC}_{29}$  as well as their frequency distributions were similar when comparing model results from different



**Fig. 6.** Violin plots of SOC changes for wheat after 29 years at reference scale showing the estimated probability density.

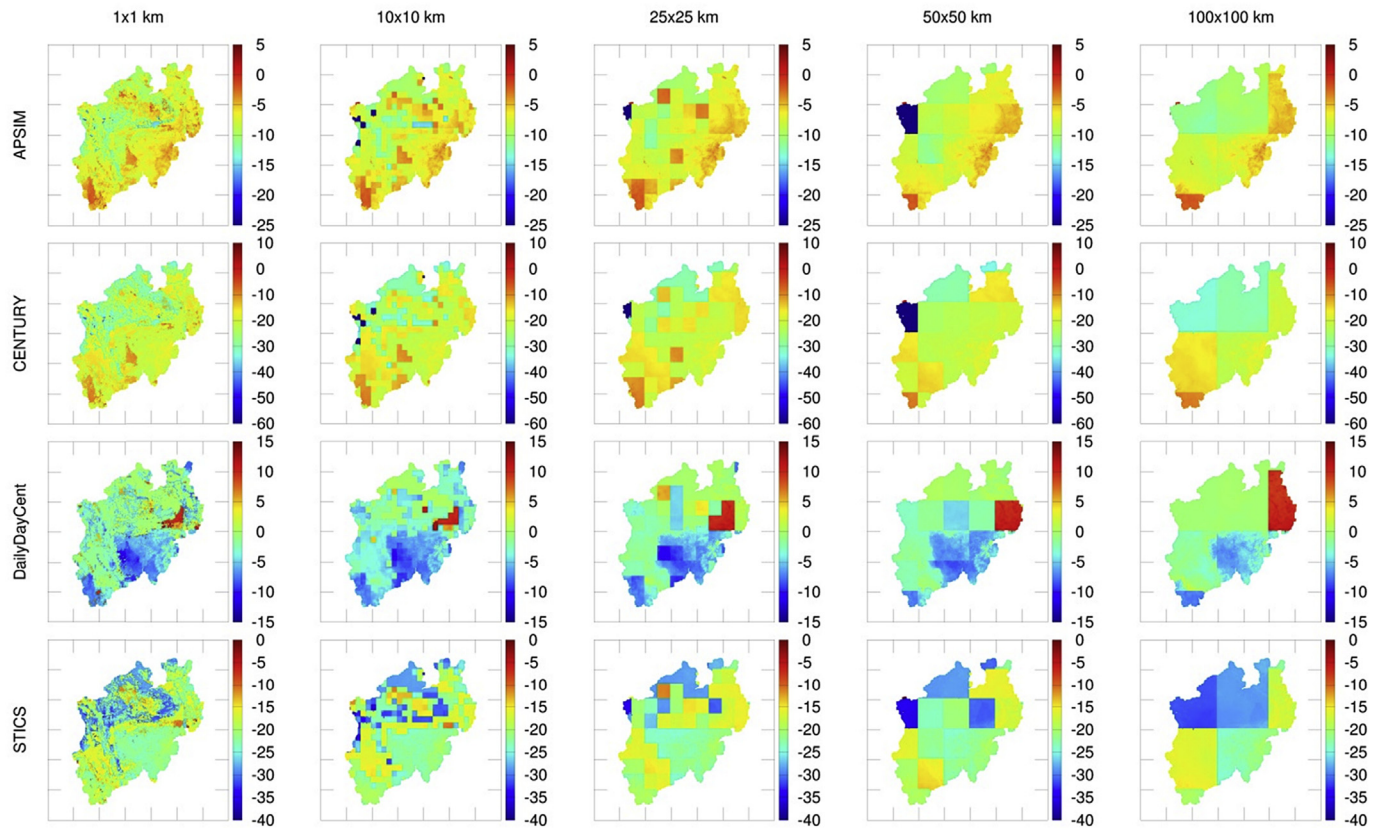


Fig. 7. Spatial distribution of modeled  $\Delta \text{SOC}_{29}$  for aggregation levels 1, 10, 25, 50 and 100 km for maize.

aggregation levels for the same model. In general, differences were larger between the models than between aggregation levels.

DailyDayCent results differed from results of other models because the DailyDayCent model runs were preceded by spin-up runs. Therefore, the mean initial SOC stock was around  $50 \text{ t C ha}^{-1}$  for the DailyDayCent run while it was around  $90 \text{ t C ha}^{-1}$  for the other models. The spatial pattern of modeled C stock changes differed from results of other models. DailyDayCent simulated larger losses in SOC in the hill ranges and smaller losses in the lowlands.

### 5.3. The effect of soil data aggregation on predicted SOC stocks and changes

For all models soil data aggregation had a pronounced effect on the spatial pattern of modeled  $\text{SOC}_t$  and  $\Delta \text{SOC}_t$  (Figs. 7 and 8). Aggregation effects expressed by the RMSE of  $\Delta \text{SOC}$  and SOC depend on the aggregation level, the scale of interest, the applied model and time. In the following we discuss exemplarily the aggregation effects after 27 years and describe the temporal pattern.

**Aggregation level:** For a scale of interest which is identical to the reference scale (1 km) model specific aggregation effects differed only slightly among aggregation levels 10, 25 and 100 km. This is consistent for all models applied. RMSE values for aggregation levels 50 km cause higher RMSE values for almost all models except DailyDayCent. As mentioned in section 5.1 the SOC stock map used for model initialization at 50 km resolution is characterized by a higher deviance to the reference SOC map than aggregation levels 10, 25 and 100 km. The observed aggregation effects for soil data resolution of 50 km are comparatively small for the model DailyDayCent because in this model application the

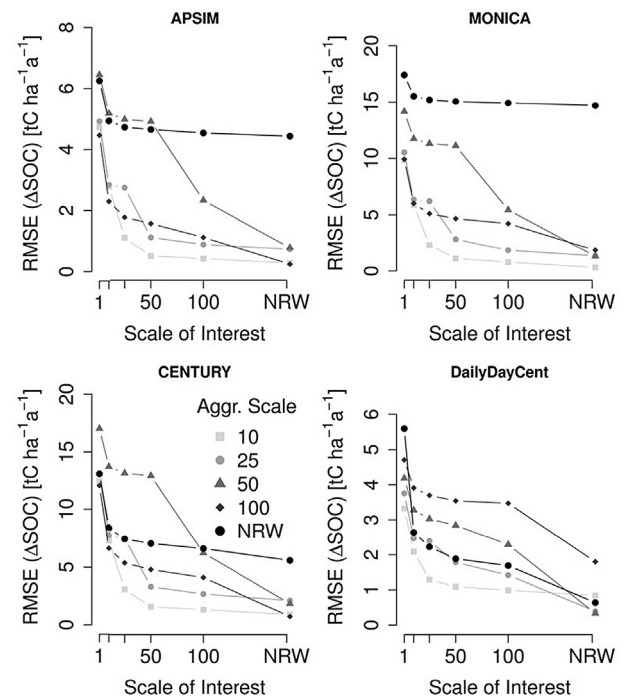


Fig. 8. RMSE ( $\Delta \text{SOC}_{29}$ ) for soil aggregation for wheat monoculture, Scale of interest (model output aggregation): 1 km, 10 km, 25 km, 50 km, 100 km and NRW; Aggregation scale (model input aggregation): 10 km (square), 25 km (circle), 50 km (triangle), 100 km (diamond), NRW (black circle).

initial SOC maps were modified by an a priori spin-up run. The representation of soil data by just one soil data set at NRW aggregation level yielded variable aggregation errors that were in general above aggregation errors of aggregation levels 10, 25 and 100 km.

**Model:** The variability of the aggregation effect ( $RMSE(\Delta SOC_t)$ ,  $RMSE(SOC_t)$ ) was higher between models than between aggregation scales within a unique model. When neglecting the aggregation level NRW the maximum range of aggregation effects ( $RMSE(\Delta SOC_{27})$ ) within a model was found for CENTURY ranging from 12.2 to 17 t C ha<sup>-1</sup> for a scale of interest of 1 km. For these aggregation levels ranges of aggregation effects between models were generally higher than 7 t C ha<sup>-1</sup> (Fig. 8). For aggregation levels 10, 25 and 100 km relative errors ( $rRMSE(\Delta SOC_{27})$ ) ranged from around 50% ( $\pm 10\%$ ) for models with high mean absolute SOC changes (APSIM, CENTURY, STICS, COUP, MONICA) to around 100% ( $\pm 20\%$ ) for model applications with small SOC changes (DailyDayCent, HERMES) as shown in Fig. 11. Relative aggregation effects  $rRMSE(\Delta SOC_{27})$  for aggregation level 50 km were around 80% for models with high mean absolute SOC changes (APSIM, CENTURY, STICS, COUP, MONICA) and between 110 and 120% for HERMES and DailyDayCent, respectively.

**Scale of interest:** Aggregation effects ( $RMSE(\Delta SOC_{27})$ ,  $RMSE(SOC_{27})$ ) and normalized aggregation effects ( $rRMSE(\Delta SOC_{27})$ ) decreased with coarser model output resolution (Fig. 8, Fig. 11, 12). In general, the decrements of aggregation effects were high for a decrease of spatial resolution of model outputs between scales of interest of 1 and 10 km but lowered beyond a scale of interest of 10 km. When plotting aggregation effects against the scale of interest related curves converged to a minimum which usually was approached at a scale of interest corresponding to NRW (Fig. 8, Fig. 11, 12). The spatial input data resolution affected the shape of the relationship between aggregation effects and scale of interest. For an input data resolution of 10 km aggregation effects decreased considerably between scales of interest of 1 km and 50 km but remained constant beyond a scale of interest of 50 km for all models. Soil data aggregation at a resolution of 25 km resulted in decreasing aggregation effects between scales of interest of 1 km and 100 km and no relevant reduction further on for all models except DailyDayCent. Here the minimum was reached at a scale of interest representing NRW. Aggregation effects decreased almost linearly with increasing scale of interest for a soil data resolution of 50 km. The effect of model output resolution based on soil input

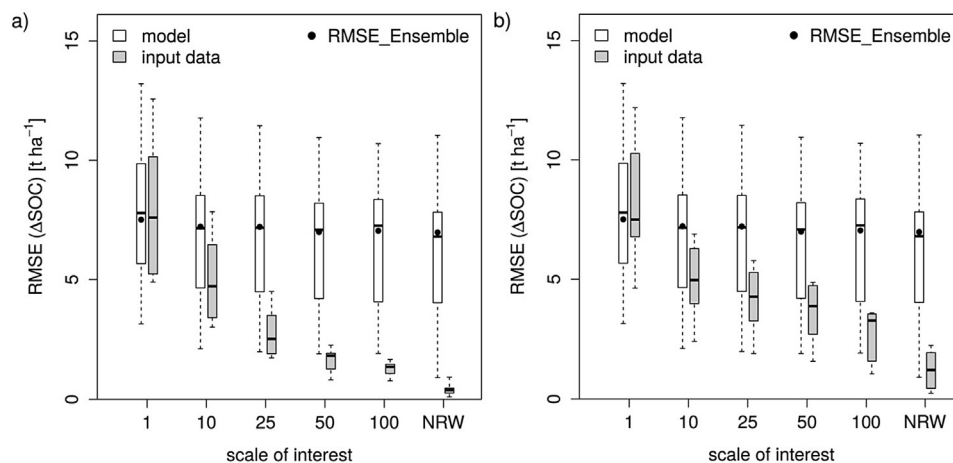
data resolution of 100 km differed between models. For models APSIM, MONICA, CENTURY, COUP and STICS decreases of aggregation effects with decreasing model output data resolution were observed that compared to soil data aggregation level of 25 km. For DailyDayCent and HERMES reductions of aggregation effects were less pronounced between scales of interest of 1 km and NRW.

**Time:** Aggregation effects ( $RMSE(\Delta SOC)$ ) increased nonlinearly with time while  $RMSE(SOC)$  decreased with time for all models except DailyDayCent where a slight increase was observed (Fig. 10). Relative aggregation effects  $rRMSE(\Delta SOC)$  increased with time for models DailyDayCent and HERMES and decreased for MONICA, COUP, APSIM and STICS. Temporal invariant relative aggregation effects were observed for CENTURY.

Variability between models was described by the  $RMSE_{model}$  of equation (9) (with the model ensemble mean as reference) and was quantified based on model simulations driven by reference data (soil properties in  $1 \times 1$  km). For a scale of interest of 1 km, the variability between models and the observed soil aggregation effects (within a model application) were in the same range. When aggregating the model output data model variability was hardly affected (Fig. 9). This is shown in Fig. 9 for those models without a priori spin-up runs (DailyDayCent not included). With increasing scale of model output aggregation, the deviation of individual models from the model ensemble mean slightly decreased for all models with  $RMSE$  reductions ranging from 0.4 t C ha<sup>-1</sup> for APSIM to 2.5 t C ha<sup>-1</sup> for COUP (for an aggregation scale of 10 km). The decrease in variability between models was marginal compared to the error reduction of aggregation effects when aggregating model output. The variability of  $\Delta SOC$  and  $SOC$  between models (equation (9)) increased with time and converged towards 7 t C ha<sup>-1</sup> for  $\Delta SOC$  and 12.5 t C ha<sup>-1</sup> for  $SOC$ , respectively.

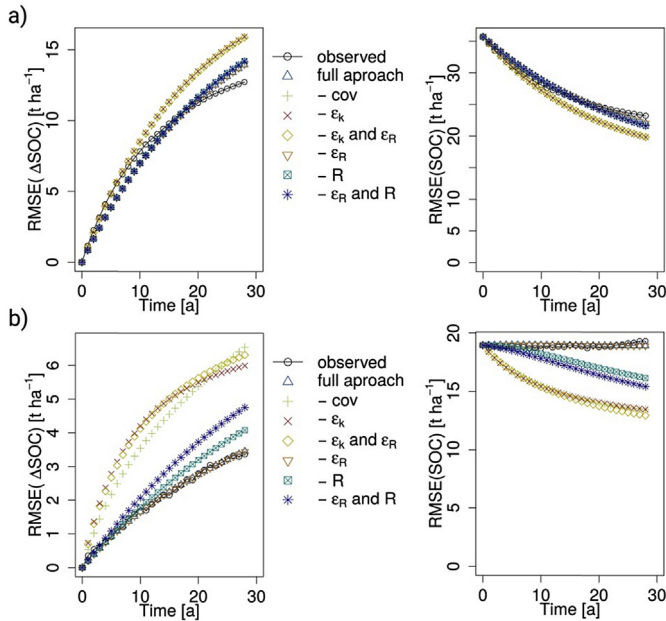
The aggregation effects expressed as the  $RMSE$  of errors in modeled SOC stocks and modeled SOC stock changes correlated well with the  $RMSE$  of initial SOC values ( $RMSE(SOC_0)$ ) which applied for all modeling periods and combinations of aggregation scales and scales of interest (Table 1). These correlations were less pronounced for the model results of DailyDayCent and HERMES whereas for the first model an a priori spin-up run was applied and the latter is characterized by relative high inert organic carbon fractions (see supplementary material A) limiting the effects of deficient initial SOC stocks.

Based on these findings the observed aggregation effects in this



**Fig. 9.** Comparison of the variability between models ( $RMSE$  ensemble) and the errors caused by soil data aggregation for different scales of interest (model output aggregation). Empty box plots represent the deviation of each individual model to the model ensemble mean. Filled box plots show the aggregation effects for individual models. Results from DailyDayCent are not included. a) input aggregation level = 10 km; b) input aggregation level = 100 km.





**Fig. 10.** Observed and estimated time series of  $RMSE(\Delta SOC_t)$  and  $RMSE(SOC_t)$  for the aggregation scale 10 km for a) the Century and b) the DailyDayCent model.

**Table 1**

Coefficients and model performance for the correlation between  $RMSE$  of initial  $SOC$  ( $RMSE(SOC_0)$ ) and aggregation errors  $RMSE(\Delta SOC_{t=27})$  and  $RMSE(SOC_{t=27})$  for a selected modeling period of 27 years.

| Model        | RMSE( $\Delta SOC_{t=27}$ )<br>versus $RMSE(SOC_{t=0})$ |                | RMSE( $SOC_{t=27}$ )<br>versus $RMSE(SOC_{t=0})$ |                |
|--------------|---------------------------------------------------------|----------------|--------------------------------------------------|----------------|
|              | slope                                                   | R <sup>2</sup> | slope                                            | R <sup>2</sup> |
| APSIM        | 0.123                                                   | 0.999          | 0.881                                            | 1.000          |
| STICS        | 0.385                                                   | 0.962          | 0.690                                            | 0.991          |
| MONICA       | 0.275                                                   | 0.987          | 0.763                                            | 0.988          |
| Century      | 0.353                                                   | 0.998          | 0.647                                            | 1.000          |
| COUP         | 0.209                                                   | 0.995          | 0.794                                            | 1.000          |
| HERMES       | 0.237                                                   | 0.652          | 0.895                                            | 0.975          |
| DailyDayCent | 0.168                                                   | 0.739          | 1.043                                            | 1.000          |

study could be estimated by equations (32) and (33):

$$RMSE(\Delta SOC_t) = \alpha_t \times RMSE(SOC_0) \quad (32)$$

$$RMSE(SOC_t) = \beta_t \times RMSE(SOC_0) \quad (33)$$

where  $t$  is the time and  $\alpha$  and  $\beta$  are coefficients that depend on the model in use and the length of the simulated period. The term  $RMSE(SOC_0)$  is the root mean squared error expressing the deviance between the initial  $SOC$  reference map and its counterparts produced by soil data aggregation. Model specific aggregation effects of different scales ( $RMSE(\Delta SOC_t)$  and  $RMSE(SOC_t)$ ) normalized by  $RMSE(SOC_0)$  vary within a narrow range. The strong relationship between aggregation effects and  $RMSE(SOC_0)$  indicate that the error of the initial  $SOC$  map is the most important driver of aggregation effects for the majority of model applications in this exercise. Thus aggregation effects on decomposition dynamics and carbon input are less relevant and the parameters  $\alpha$  and  $\beta$  of equations (32) and (33) might be explained by terms of equations (30) and (31). This is examined in section 5.4.

#### 5.4. Approximation of aggregation effects based on SICA

The estimation of soil data aggregation effects based on equation (29) requires comprehensive data on model parameters and model initialization variables ( $SOC_0$ ). Aggregation effects for these variables and covariances between them have to be known. In section 4.4, we derived possible simplifications with reduced demands on required information. Because the information on soil properties in high resolution is often restricted simplified approaches with small data requirements might be better transferable to other regions and up-scaling issues than approaches with high data demands. Therefore, several simplifications of equation (29) were compared with  $RMSE(SOC_t)$  and  $RMSE(\Delta SOC)$  directly quantified from model output.

Based on cell specific determined parameters,  $SOC$  initialization and carbon input we applied equation (29) and several simplifications to approximate the aggregation effect. Following simplifications were tested:

1. Full approach: Equation (29) applies, e. g.  $k$ ,  $R$ ,  $fract_{IOM}$ ,  $SOC_0$  and their deviances from the high resolution reference data have to be known. Covariance between variables is considered.
2. all covariances: Parameters  $k$ ,  $R$ ,  $fract_{IOM}$ ,  $SOC_0$  and their deviances from the high resolution reference data have to be known. Covariance between data is not considered.
3.  $\epsilon_k$ : The parameter  $\epsilon_k$  has no relevance
4.  $\epsilon_R$ : The parameter  $\epsilon_R$  has no relevance
5.  $\epsilon_k$  and  $\epsilon_R$ : The parameters  $\epsilon_k$  and  $\epsilon_R$  have no relevance. Except of  $SOC_0$  no input from the reference scale is required.
6.  $R$ : Differences in decomposition dynamics of added material by different decomposition rates in the reference scale and aggregation scale does not matter
7.  $\epsilon_R$  and  $R$ : Differences in carbon input rates and decomposition of carbon input are irrelevant

The Nash- Sutcliffe index (NSI) as well as the minimum and maximum residue were quantified to approximate the deviance of observed and estimated time series of aggregation effects  $RMSE(\Delta SOC_t)$  and  $RMSE(SOC_t)$  for each model and each quantification method (Table J1).

The full equation (29) resulted in good estimations of observed aggregation effects for all models. Low and negative NSI for estimated  $RMSE(SOC_t)$  were observed for the models HERMES and DailyDayCent. For these models the variance of observed  $RMSE(-SOC_t)$  was very low (Fig. 10d). The NSI normalizes the quantified deviations by the variance of observations and so even negative NSI might occur although the visual check of the model fit might be acceptable (Fig. 10d). Deviances between observed and modeled aggregation effects mainly result from the limited flexibility of the calibration problem (Locally, only the decomposition rate was calibrated).

Good approximations of the aggregation effect could also be derived if the influence of  $\epsilon_R$  or  $R$  were neglected. All other simplifications resulted in discrepancies between observation and estimation. In general, discrepancies increased in the order full approach  $< \epsilon_R - \epsilon_R < -R < -R$  and  $\epsilon_R < -$  all covariances  $< -\epsilon_k < -\epsilon_k$  and  $\epsilon_R$ . Model fits of simplifications were acceptable for all models that used the original initialization data and modeled pronounced  $SOC$  losses during the simulation period (CENTURY, MONICA, COUP). Only the simplifications  $-\epsilon_R$ ,  $-R$ ,  $-R$  and  $\epsilon_R$  produced acceptable model fits of  $RMSE(\Delta SOC)$  for the remaining models. Out of these three approaches  $-R$ ,  $-R$  and  $\epsilon_R$  failed to estimate reliable time series of  $RMSE(SOC)$  for DailyDayCent simulations.



## 6. Discussion

### 6.1. Controls on soil data aggregation effects

Just as models are used as simplified representations of delimited systems in nature we used a simplified approach (SICA) to describe dynamics, sensitivities and interaction of processes controlling SOC fluxes of biogeochemical process models. SICA is a compartment model and describes soil organic matter decomposition by first order kinetics. These mathematical structures are shared by most of soil organic matter models developed since 1930 (Sierra et al., 2012) including participating models. In dependence of number and linkage of defined SOC pools the degree of nonlinearity is variable between these models (Manzoni and Porporato, 2009) and depends on the distribution of SOC between defined SOC pools beyond effects of temperature, soil moisture, texture and nitrogen. SICA has just one degradable SOC pool and decomposition kinetics were fitted against modeled SOC trends of participating models therefore decomposition dynamics of SICA integrate also effects of initial SOC distribution among SOC pools of the respective model. For this reason model specific calibrated parameters of SICA are only valid for the realized initialization scheme, range of environmental conditions and the temporal horizon of the modeling exercise.

We used approaches based on the analytical solution of SICA to describe aggregation effects on simulated  $\Delta$ SOC and SOC series. The developed approaches assume that environmental and management conditions affecting SOC are static, restricting the application of equation (29) to problems without trends or high temporal variation in management and natural conditions. Altered SOC stocks might feedback on crop growth and soil physical properties, which is not considered by SICA. However, in our modeling example N limitation should be less relevant given the assumed nitrogen supply of 208 and 238 kg N ha<sup>-1</sup> a<sup>-1</sup> for wheat and maize, respectively and the effect of SOC changes on soil physical conditions is not accounted for in participating models.

The underlying experimental data was produced by model up-scaling using 7 different biogeochemical models for two artificial mono cropping systems. The simulated trends in SOC development were negative for all models. We suppose that these modeled SOC losses are caused by the mono-cropping systems with low annual plant residue input and neglected organic amendments. Additionally, the modeling study assumes original grassland and forest areas to be covered by crops (wheat or maize). However, similarly pronounced decreases of SOC stocks in cropland soils were reported for other European regions and European flux tower measurement networks too (Bellamy et al., 2005; Goidts and van Wesemael, 2007; Kutsch et al., 2010). Profound changes of mean soil organic contents in the top soil of arable land in NRW are reported by Steinmann et al. (2016) too. Averaged soil organic carbon contents increased between 1979 and 1988 from 1.87 to 2.82% and then decreased to 1.42% in 2015. Steinmann et al. (2016) explain these pronounced SOC changes by cropland to grassland conversion between 1980 and 1990 and dominating grassland to cropland conversions between 1988 and 2015. In our experiment the whole area of North Rhine-Westphalia was used for modeling. Former forest and grassland areas which are in general characterized by elevated SOC stocks (Poeplau et al., 2011) compared to croplands were included. Therefore the underlying initial SOC data spanned a wide range of SOC contents from 0.01% till around 15%. This variability influenced also the aggregation effect on initial SOC stocks ranging from an RMSE(SOC<sub>0</sub>) of 37–41 t C ha<sup>-1</sup>.

According to SICA based approaches (equations (17) and (18)) aggregation effects per grid cell result from the sum of 4 terms expressing the contribution of SOC<sub>0</sub> (SOC<sub>0</sub> ×  $f_1(t, k, \epsilon_k, fract_{IOM})$ ),

$\Delta$ SOC<sub>0</sub> ( $\Delta$ SOC<sub>0</sub> ×  $f_2(t, k, \epsilon_k, fract_{IOM})$ ),  $R(R \times f_3(t, k, \epsilon_k))$  and  $\Delta R(\Delta R \times f_4(t, k, \epsilon_k))$ .

Aggregation effects are linear correlated to absolute initial SOC whereas the slope of this correlation is determined by implied decomposition dynamics, aggregation effects on implied decomposition dynamics and time. The effect of absolute initial SOC stock decomposition by deficient decomposition dynamics decreases with time. Aggregation effects driven by absolute SOC stocks are amplified by slow implied decomposition dynamics ( $\epsilon_k = \text{const.}$ ) so these aggregation effects are more relevant in areas with accumulated C stocks caused by temperature or soil moisture limitations. The contribution of the mentioned interaction of initial SOC stock and deficient decomposition dynamics on aggregation effects was relevant for all models. This is indicated by the comparison of simplified approaches used to quantify aggregation effects in section 5.4. The fit between observed and estimated aggregation effects for the models APSIM, STICS and DailyDayCent decreased strongly when assuming that the deficient decomposition of active SOC does not contribute to the aggregation effects (Table J1). Effects were moderate for the remaining models. For the up-scaling experiment mean initial SOC stocks of NRW with 90 t C ha<sup>-1</sup> were high compared to averaged cropland SOC stocks of other regions in Germany, France and Belgium ranging from 50 t C ha<sup>-1</sup> to 58 t C ha<sup>-1</sup> (Wiesmeier et al., 2013; Martin et al., 2011; Lettens et al., 2004). Therefore we assume that aggregation effects on modeled SOC changes and stocks caused by the interaction of initial SOC stocks and deficient decomposition dynamics in this study are comparably high compared to other regions of temperate Europe.

According to equations (17) and (18) aggregation effects of estimated trends in SOC and  $\Delta$ SOC are linearly correlated to errors of the initial SOC ( $\Delta E_{SOC_0}$ ). Decomposition dynamics and related errors were less relevant for aggregation effects caused by deficient initial SOC at equilibrium but controlled its temporal dynamics before equilibrium was established. Errors of the initial SOC map were relevant for observed aggregation effects for all models shown by the strong correlation between aggregation effects and RMSE(SOC<sub>t</sub>) and RMSE( $\Delta$ SOC<sub>t</sub>), respectively. The relative contribution of aggregation effects by deficient SOC<sub>0</sub> data was driven by the aggregation effects on SOC<sub>0</sub> which depends on the spatial distribution and variability of SOC at reference scale. Aggregation effects on initial SOC had a strong impact on errors of simulated SOC changes for models whose decomposition dynamics were less sensitive to soil data aggregation (CENTURY, COUP, MONICA).

Aggregation effects from carbon inputs and deficient carbon inputs were determined by decomposition dynamics ( $k$ ) and its aggregation errors ( $\epsilon_k$ ) with higher prediction errors of  $\Delta$ SOC<sub>t</sub> and SOC<sub>t</sub> for decreasing  $k$ . Aggregation effects from carbon inputs were most relevant for DailyDayCent and HERMES but had only marginal effects for the remaining models.

### 6.2. Requirements on aggregated soil input data for SOC up-scaling

Model errors decreased with model output data aggregation thereby RMSE(SOC<sub>t</sub>) and RMSE( $\Delta$ SOC<sub>t</sub>) were correlated with RMSE(SOC<sub>0</sub>), so the error of the initial SOC map was of outstanding importance for the aggregation effect. The aggregation scheme used which is based on the most common soil type per aggregation grid cell can produce locally biased SOC<sub>0</sub> maps and thus does not necessarily minimize the error of initial SOC information. This is also found in this exercise. Especially, for soil data aggregation to a spatial resolution of 50 km grid cell size initial SOC data information was locally biased causing higher aggregation effects for SOC model results than soil data aggregation to a coarser resolution of 100 km. The possibility of biased SOC data by majority based soil data aggregation follows also from a thought experiment where we

assume evenly distributed principal and secondary soil types (each characterized by a specific SOC) across the region of interest. Here aggregation by majority can completely extinguish the information of secondary soil types and will cause biased model results for the whole region of interest even for spatially aggregated output data.

In this study, we developed an approach to quantify soil data aggregation effects based on the simple soil carbon model SICA which was calibrated to mimic dynamics and sensitivities of applied biogeochemical models. It was demonstrated that beyond the aggregation errors of initial SOC maps, the aggregation of soil properties that control decomposition dynamics and the carbon input play a role for certain circumstances related to soil properties and the model applied. Moreover, the covariances between these factors determine the error caused by soil data aggregation as was seen in section 5.4 where we compared error estimation with and without consideration of covariances. Therefore, soil data aggregation (and soil data in general) has to ensure the consistency between the initial SOC and those soil properties controlling the temporal development of water content, soil temperature and factors influencing simulated crop growth and predicted decomposition dynamics. Contrary to the linear relationship between modeled SOC and initial SOC (equation (13)), the relationship between modeled SOC and the decomposition rate  $k$  is nonlinear, so up-scaling errors are likely to occur when soil data aggregation is done by averaging instead of soil data aggregation by majority. Moreover, the covariance between  $k$  related terms and the initial SOC map might be compromised by averaging soil properties.

For up-scaling of nonlinear biogeochemical models, Heuvelink and Pebesma (1999) recommend the consideration of the point support of biogeochemical models. Model runs have to be done on a representative and well described set of sampling points first, before areal model output information is derived by aggregation or geostatistical methods. This sampling procedure will avoid up-scaling errors that arise from nonlinear model responses. This procedure is also applied by Ogle et al. (2010) who estimate regional and national SOC trends for agricultural soils in the US based on model applications on a set of sites spread across the region of interest and a stratified up-scaling of model results. In our study, soil data aggregation effects can be reasonably explained by errors of the initial SOC map for most models because aggregation effects on initial SOC overrode aggregation effects on decomposition dynamics and carbon input from crop residues. However, the effect of these factors increase with decreasing error of the initial SOC map shown by the model runs of DailyDayCent. In the same time, the relative error of modeled SOC changes was biggest for these models. Therefore, errors might be avoided if up-scaling studies are based on model applications on a representative set of sampling points with consistent sets of soil properties instead of aggregated soil data. Nevertheless, there are good reasons to perform up-scaling studies with block support (areal representation of model input data), because some model input information like management data from regional statistics or agro-economic models (scenarios) is often only available with block support (Ewert et al., 2011). For these cases, soil data aggregation effects on modeled SOC and  $\Delta$ SOC can be huge at fine model output resolution (with  $\text{RMSE}(\Delta\text{SOC}_{27})$  ranging between 50% and 120% of mean modeled SOC changes in this study) and coarse model output resolution (with  $\text{RMSE}(\Delta\text{SOC}_{27})$  ranging between 3% and 50% of mean modeled SOC changes in this study). For all models we observed a continuous decrease of aggregation effects with increasing scale of interest for an input data resolution of 10, 25, 50 and 100 km. In our modeling experiment soil input data resolutions of 10 and 25 km resulted in minimized errors at a model output data resolution of 50 km, 100 km and NRW, respectively. An adequate aggregation of soil data must consider both, the effects of deficient initial SOC and

the effects (errors) of other aggregated soil properties that control decomposition dynamics and carbon input of the respective model. The aggregated soil data of NRW showed regional biased data for  $\text{SOC}_0$  at a spatial resolution of 50 km while texture and geo-hydrological properties were regionally biased at spatial resolution of 100 km (Fig. 5). Models driven by the original SOC data and characterized by a pronounced decay of SOC showed the strongest aggregation effects for a soil data resolution of 50 km (Fig. 8, APSIM, MONICA, CENTURY) across all scales of interest. Models driven by modified initial SOC data and models with high fractions of inert pools showed highest aggregation effect for a soil data resolution of 100 km. In most cases, the relative importance of specific soil properties and their errors by spatial aggregation is not a priori predictable therefore up-scaling studies with focus on SOC changes should be based on the highest feasible spatial resolution. However, computation time becomes relevant with high resolution input data. For the finest spatial resolution of 1 km (36116 cells) computation time ranged between half an hour and 2 weeks. The number of necessary model runs decreases with the square of scaled grid cell size ( $\text{time}_{\text{agg\_input data}}/\text{time}_{\text{ref\_data}} = 1/(\text{scaled\_gridcell\_size})^2$ ), so input data at 10 km resolution would be around hundred times faster (0.5 min–3.5 h in our example). This relation highlights the need for balancing requirements on computation time, output data resolution and prediction errors.

### 6.3. Soil data aggregation effects in relation to model variability

In this exercise, soil data aggregation effects decreased strongly by output data aggregation. Output data aggregation only marginally leveled out differences of model outputs between models showing that errors caused by deficient model structure are likely to persist even for low demands on spatial model output resolution. Moreover, it demonstrates that methods (Gaussian error propagation or Monte Carlo sampling) used to quantify the uncertainty of spatial aggregated model results are not applicable if they neglect the spatial correlation of structural model errors.

Differences in the initialization of SOC pools per model are one likely reason for the variability between model results. In this study, the distribution of SOC between pools was initialized by applying default fractions with chosen fractionation schemes approximating equilibrium conditions for cropland use (except DailyDayCent). Spin-up runs for each computational grid cell were not performed, because this procedure would have compromised the spatial initial SOC distribution and initial SOC error distribution for the different aggregation levels like it was observed for DailyDayCent spin-up runs. The variability between models might have been limited by harmonizing initialization schemes for participating models. Carvalhais et al. (2008) and Hashimoto et al. (2011) describe initialization techniques called fast and slow relaxation schemes. These techniques preserve the initial SOC stock after spin-up by scaling methods and would have been an option to harmonize model initialization and decrease the variability between models. However, the fast relaxation scheme can produce unstable model behavior and both methods were developed for forest ecosystems and need further testing for varying conditions and models (Hashimoto et al., 2011). Compared to forests, agricultural systems are characterized by short duration management changes questioning the assumption of equilibrium conditions for model initialization. This was shown by Foreid et al. (2012) who compared model scenarios on sites across England and Wales initialized with the fast relaxation method at the one hand and on initial SOC distributions among SOC pools calibrated on observed SOC changes at the other hand. Interestingly, both initialization methods showed pronounced site specific differences but compared well for spatially aggregated predictions of SOC changes.

The results of this study confirm the observations of Ogle et al. (2010) that model uncertainties dominate the uncertainty of spatial aggregated results for SOC trend simulations. While Ogle et al. (2010) used stochastic approaches to describe the model uncertainty of a single model, uncertainty estimates in this study were based on the variability between 6 models. Because measured SOC changes were not available in this study the bias of the model ensemble was not considered. However, based on equation (9) the bias related error of the model ensemble has to be positive (or zero at best) and so does not level out with errors by variability between models. Therefore, quantified errors caused by model variability mark the lower limit of real model errors that integrate bias and variance of the model ensemble.

The results of this study have shown that model related errors are of outstanding relevance for SOC up-scaling studies. The relative influence of model errors increased with spatial output data aggregation compared with soil data aggregation errors (Fig. 9). Therefore, the need for well calibrated and initialized SOC models has to be stressed in the context of model up-scaling. Ogle et al. (2010) used measured SOC trends of an extensive data set of sampling points representing the region of interest to evaluate up-scaling results. Comparable data sets do not exist for the region of interest NRW. All of the applied models have been proven to reproduce SOC trends of field data (Supplementary material A). However, methods on the transfer of model uncertainty estimation from one region (coincidence of weather, soil and management) to another region do rarely exist. Based on the observed variability between models we conclude that SOC up-scaling studies should rather be based on model ensemble runs than on the application of a single model. Especially, the uncertainty of model outputs at coarse resolution might be underestimated without an adequate spatial aggregation scheme of model structural uncertainty.

#### 6.4. General methods to approximate soil data aggregation effects

We aimed to study soil data aggregation effects on SOC up-scaling to investigate the requirements of aggregated soil input data and the sensitivity of aggregation errors on applied models, time, aggregation level and specific soil properties. Like in numerous studies analyzing aggregation effects (Hoffmann et al., 2016a) the results of this study are based on a subdomain of possible coincidences of models, model input and aggregation levels therefore the transferability of observed aggregation effects of this modeling study to other regions and scenarios is restricted. We derived and tested tools that might also be applicable beyond the constraints of this modeling study. Based on SICA we formulated equations that express aggregation effects as functions of system properties (spatial distribution of SOC, carbon input), SOC-sub-model properties (decomposition dynamics expressed by  $k$ , inert SOC fraction, fraction of carbon input that enters SOC) and the spatial distribution of aggregation effects on initial SOC, decomposition kinetics and carbon input (equation (28)).

These approaches proved to estimate observed temporal development of aggregation effects for the investigated subdomain well. However, for an approximation of aggregation effects knowledge of model parameters, grid cell specific initial SOC, decomposition dynamics, carbon input and aggregation errors of these factors would be required. In this exercise these data has been derived by model runs on reference scale and aggregation scale. An ex ante approximation of aggregation effects with this approach is not possible unless there are other methods to estimate the spatial distribution of carbon input from crop residues and decomposition dynamics as well as their errors caused by soil data aggregation. In dependence on the specific biogeochemical model, parameters that determine decomposition kinetics, like the decomposition rate or

the carbon use efficiency are modulated by temperature, soil moisture, texture or N availability (Supplementary material A). The spatial distribution of these factors or related proxies might serve as a basis to describe the spatial distribution of implied decomposition dynamics and the related errors for specific models by stochastic approaches. Hoffmann et al. (2016a) found that aggregation effects of crop yields could be described by the AWC (available water capacity) while Kuhnert et al. (2016) examined weather data aggregation effects on net primary productivity (NPP). Ongoing research in this field might lead to an approach to describe aggregation effects on carbon input quantification from crop residues.

In our model experiment, the effect of aggregated initial SOC data strongly influenced the aggregation effects and overrode effects from aggregation errors of carbon input and decomposition dynamics for some models that were run on the original aggregated soil data. For these conditions a simplified approach proved to describe the temporal development of aggregation effects adequately. The coefficients of these regression equations could be derived by model parameters of SICA and time. Parameters required encompass the implicit decomposition rate, the IOM fraction and the RMSE (SOC<sub>0</sub>). All parameters can be estimated by model applications on aggregated data except for the error of aggregated SOC<sub>0</sub> data. Knowledge of the error of the initial SOC map is essential for the approximation of aggregation effects on SOC<sub>t</sub> and  $\Delta$ SOC<sub>t</sub> up-scaling. This follows from derived equations (29)–(31) and from the observed strong correlation between RMSE(SOC<sub>0</sub>) and RMSE(SOC<sub>t</sub>). This error depends on geostatistical properties of SOC<sub>0</sub> at reference scale (e.g. spatial variability and autocorrelation) and the implementation of soil data aggregation (resolution and method).

## 7. Conclusion

Soil data aggregation had relevant effects on modeled SOC changes and SOC stocks for a modeling study with 7 biogeochemical models that were applied for the region North Rhine-Westphalia assuming mono cropping systems of maize and wheat. In our modeling study aggregation effects were between 50 and 120% of mean modeled SOC stock changes at 1 km resolution when soil data was aggregated by majority to grids of 10 km, 25 km, 50 km, 100 km and NRW (one soil for NRW) resolution. The effect of input data aggregation decreases with coarser model output resolution.

Soil data aggregation affected the spatial representation of initial SOC, SOC decomposition dynamics of participating models and the carbon input. These factors and their related aggregation effects controlled aggregation effects of simulated SOC<sub>t</sub> and  $\Delta$ SOC<sub>t</sub>. Out of these factors the aggregation effects on initial SOC were most important for aggregation effects on modeled SOC and SOC changes followed by aggregation effects on decomposition dynamics and carbon input. The relative contribution of these factors varied in dependence on the model in use, the method of model initialization, the amplitude of named factors and their covariance.

Output data aggregation only marginally reduced differences of model outputs between models indicating that errors caused by deficient model structure are likely to persist even if requirements on the spatial resolution of model outputs are low. Methods (Gaussian error propagation or Monte Carlo sampling) used to quantify the uncertainty of spatial aggregated model results are not applicable if they neglect the spatial correlation of structural model errors.

## Author contributions

Conceived and designed the experiments: HH BG RD GZ JC HR



DW EC EL HE XS KCK CN MK JY EH ET MB GT MM LD PPR ZZ EW FT RR BK DC SA LW SS TG FE. Performed the experiments: BG RD HH GZ JC HR EC EL HE XS KCK CN MK JY EH ET ZZ EW BK DC. Analyzed the data: BG RD SG. Wrote the paper: BG RD.

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## Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.envsoft.2017.06.046>.

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